

## **X-ray Diffraction and Microscopy**

### **Syllabus:**

Properties of x-rays: Continuous and characteristics x-rays, absorption, filter, production and detection of x-rays – [4 hours](#)

Diffraction of x-rays: special topics on crystallography, directions and intensities of diffracted beams and diffraction under non ideal condition (Scherrer formula) – [6 hours](#)

Applications: Determination of crystal structures of polycrystalline materials, precise lattice parameter measurement, crystallite size and lattice strain measurement, quantitative phase analysis and residual stress measurement - [8 hours](#)

Optical Microscopy: Fundamentals of metallurgical microscope, imaging and quantitative metallography – [3 hours](#)

Scanning Electron Microscopy: Specimen beam interaction, construction of scanning electron microscope (SEM), principles of image information, energy dispersive x-ray spectroscopy (EDS) and wavelength dispersive x-ray spectroscopy (WDS) - [6 hours](#)

Transmission Electron Microscopy: Introduction to construction of transmission electron microscope (TEM), principles of image formation, interpretation of diffraction pattern and sample preparation techniques – [11 hours](#)

Fundamentals of scanning tunneling microscopy (STM) and atomic force microscopy (AFM) - [4 hours](#)